IGCSE 0607 Practice Questions — E1.3 — Combined Questions & Markscheme

Powers and roots

Total Questions: 18 — **Total Marks:** 94 Marks

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  → **Paper 2 (Non-Calculator)** *(12 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

1 **(a)** Write down the value of $\sqrt{121}$.

..... **[1]**

(b) Work out the value of

$$4^3 - \sqrt[3]{27}$$

..... **[2]**

2 **(a)** Work out the value of

$$2^3 \times \sqrt{81}$$

..... **[2]**

(b) Write down the value of $\sqrt[3]{125}$.

..... **[1]**

3 **(a)** Work out the value of

$$\sqrt{169} + 2^3$$

..... **[2]**

(b) Work out the value of

$$4^2 \times \sqrt[3]{125}$$

..... **[2]**

4 **(a)** Work out the value of

$3^3 - \sqrt{144}$

..... **[2]**

(b) Work out the value of

$2^4 \times \sqrt[3]{125}$

..... **[2]**

5 **(a)** Write down the value of

$\sqrt{196}$

..... **[1]**

(b) Work out the value of

$4^3 - \sqrt[3]{125}$

..... **[2]**

(c) Calculate the exact value of

$\sqrt{13^2 - 12^2}$

..... **[2]**

6 **(a)** Work out the value of

$$5^2 - \sqrt{121}$$

..... **[1]**

(b) Work out the value of

$$\sqrt[3]{27} \times 2^4$$

..... **[2]**

(c) Work out the exact value of

$$\frac{6^2 + \sqrt{64}}{\sqrt[3]{125}}$$

..... **[2]**

7 **(a)** Work out the exact value of

$$\frac{5^2 + \sqrt{144}}{2^3 - \sqrt[3]{1}}$$

..... **[2]**

(b) Find the value of k when

$$3^k = \sqrt{81} \times 3^2$$

$k =$ ..... **[2]**

8 **(a)** Calculate the value of

$$\frac{4^3 + \sqrt{225}}{\sqrt[3]{64}}$$

Give your answer as a decimal.

..... **[3]**

(b) Given that $p = 5$ and $q = -2$, work out the value of

$$p^2 - q^3$$

..... **[2]**

9 **(a)** A cube has a volume of 125 cm^3.

Find the length of one edge of the cube.

..... cm **[1]**

(b) Work out the total surface area of this cube.

..... cm^2 **[2]**

(c) Given that $\sqrt{N} = 3^3 - 4^2$, find the value of N.

$N =$ ..... **[2]**

10 **(a)** Work out the exact value of

$$\left(\frac{\sqrt{225}+\sqrt[3]{1000}}{5^2}\right)^{-2}$$

..... **[3]**

(b) Solve the equation

$$x^{\frac{2}{3}}=\frac{\sqrt{144}}{3}$$

$x =$ ..... **[3]**

11 Solve the following equations.

(a)

$\sqrt[3]{3x+4} = 4$

$x =$ ..... **[3]**

(b)

$x^2 = 4^3 + 6^2$

$x =$ ..... , or $x =$ ..... **[2]**

(c)

$\sqrt{x} + 2\sqrt{x} = 21$

$x =$ ..... **[2]**

12 **(a)** Find the integer k such that

$$\sqrt[3]{k} = 4^2 - 3^2 - 2$$

$k =$ ..... **[2]**

(b) Calculate the exact value of

$$\frac{\sqrt{10^2 - 6^2}}{\sqrt[3]{125} - 3^2}$$

..... **[3]**

(c) Given that $y > 0$ and $\sqrt[3]{y} + \sqrt[3]{y} = 6$, work out the value of y^2.

..... **[3]**

13 **(a)** Calculate the value of

$$\frac{\sqrt[3]{450} + 12.5^2}{3.4^3 - \sqrt{80}}$$

Write down all the figures on your calculator display.

..... **[2]**

(b) Write your answer to part (a) correct to 3 decimal places.

..... **[1]**

(c) Find the largest integer n such that $n^4 < 2000$.

$n =$ ..... **[2]**

14 (a) (i) Calculate the value of

$$\sqrt[3]{8.5^2 - 4.1^3}$$

Write down your full calculator display.

..... **[2]**

(ii) Write your answer to part (a)(i) correct to 3 significant figures.

..... **[1]**

(b) Calculate the exact value of

$$\frac{4.8^2}{\sqrt{0.0576}}$$

..... **[2]**

15 **(a)** Calculate the value of

$$\frac{\sqrt{15.2^2 - 8.9^2}}{3.1^3}$$

..... **[2]**

(b) Given that $p = 4.2 \times 10^5$ and $q = 8.1 \times 10^{-3}$, work out the value of $\sqrt[3]{p \times q}$.
Give your answer correct to 3 significant figures.

..... **[3]**

16 **(a)** Calculate the value of

$$\frac{\sqrt[3]{45.6} + 2.1^4}{3.8^2}$$ Give your answer correct to 3 significant figures.

..... **[2]**

(b) A solid cube has a volume of 4913 cm^3.
Calculate the total surface area of the cube.

..... cm^2 **[3]**

17 **(a)** Calculate the value of

$\sqrt[5]{7.59375}$

..... **[1]**

(b) Find the two possible values of x in the equation

$2x^4 - 17 = 1233$

$x =$ ..... , $x =$ ..... **[3]**

(c) It is given that

$$y = \sqrt{\frac{3a^2 + b}{c}}$$

Work out the value of y when $a = 4.2$, $b = -5.7$ and $c = 1.8$.
Give your answer correct to 2 decimal places.

$y =$ ..... **[3]**

18 **(a)** Calculate the value of

$$\frac{\sqrt[4]{150000}}{4.5^3 - \sqrt{75}}$$ Give your answer correct to 3 decimal places.

..... **[2]**

(b) x is a positive integer such that $\sqrt[3]{x^2} = 16$.
Find the value of x.

$x =$ ..... **[3]**

(c) The volume of a sphere is equal to the volume of a cube with edge length 12 cm.
Calculate the radius of the sphere.
The volume, V, of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.

..... cm **[3]**

1 **(a)** 11 **[B1]**

(b) 61 **[B2]**

M1 for 64 or 3 seen.

[Back to Question] *[Total Marks: 3]*

2 **(a)** 8×9 **[M1]**

72 **[A1]**

(b) 5 **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** 21 **[B2]**

M1 for 13 or 8

(b) 80 **[B2]**

M1 for 16 or 5

[Back to Question] *[Total Marks: 4]*

4 **(a)** Calculates $3^3 = 27$ and $\sqrt{144} = 12$ **[M1]**

15 **[A1]**

(b) Calculates $2^4 = 16$ and $\sqrt[3]{125} = 5$ **[M1]**

80 **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** 14 **[B1]**

(b) 59 **[B2]**

If 0 scored, M1 for 64 or 5 seen **[M1]**

(c) 5 **[B2]**

If 0 scored, M1 for $169 - 144$ or 25 seen **[M1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** 14 **[B1]**

(b) 48 **[B2]**

If 0 scored, allow [M1] for correctly evaluating either $\sqrt[3]{27} = 3$ or $2^4 = 16$

(c) 8.8 or $\frac{44}{5}$ **[B2]**

If 0 scored, allow [M1] for a correct numerator of 44 or a correct denominator of 5

[Back to Question] *[Total Marks: 5]*

7 **(a)** $\frac{37}{7}$ or $5\frac{2}{7}$ **[B2]**

M1 for $25 + 12$ or $8 - 1$ seen.

(b) 4 **[B2]**

M1 for 9×9 or $3^2 \times 3^2$ or 81 seen.

[Back to Question] *[Total Marks: 4]*

8 **(a)** Numerator correctly evaluated to $64 + 15 = 79$ **[B1]**

Denominator correctly evaluated to 4 **[B1]**

19.75 **[B1]**

(b) Substitution leading to $25 - (-8)$ **[M1]**

33 **[A1]**

[Back to Question] *[Total Marks: 5]*

9 **(a)** 5 **[B1]**

(b) 150 **[B2]**

If 0 scored, allow [M1] for 6×5^2 or 6×25

(c) 121 **[B2]**

If 0 scored, allow [M1] for evaluating $3^3 - 4^2 = 11$

[Back to Question] *[Total Marks: 5]*

10 **(a)** $\frac{15 + 10}{25}$ **[M1]**

$\left(\frac{25}{25}\right)^{-2}$ or 1^{-2} **[M1]**

1 **[A1]**

(b) $x^{\frac{2}{3}} = 4$ **[M1]**

$x = 4^{\frac{3}{2}}$ or $(\sqrt{4})^3$ or $x^2 = 4^3 = 64$ **[M1]**

8 **[A1]**

[Back to Question] *[Total Marks: 6]*

11 **(a)** $3x + 4 = 64$ **[M1]**

$3x = 60$ **[M1]**

20 **[A1]**

(b) $x^2 = 64 + 36 = 100$ **[M1]**

$10, -10$ **[A1]**

(c) $3\sqrt{x} = 21$ or $\sqrt{x} = 7$ **[M1]**

49 **[A1]**

[Back to Question] *[Total Marks: 7]*

12 (a) 125 **[B2]**

If 0 scored, M1 for evaluating $16 - 9 - 2$ to get 5 **[M1]**

(b) -2 **[B3]**

M1 for $\sqrt{64}$ or 8 **[M1]**

M1 for $5 - 9$ or -4 **[M1]**

(c) 729 **[B3]**

M1 for $2\sqrt[3]{y} = 6$ or $\sqrt[3]{y} = 3$ **[M1]**

M1 for $y = 27$ **[M1]**

[Back to Question] *[Total Marks: 8]*

13 (a) 5.3990299... **[B2]**

M1 for numerator evaluated to 163.91... or denominator evaluated to 30.359...

(b) 5.399 **[B1]**

(c) 6 **[B2]**

M1 for evaluating $6^4 = 1296$ and $7^4 = 2401$

[Back to Question] *[Total Marks: 5]*

14 (a) (i) 1.49315... **[B2]**

If 0 scored, M1 for 3.329 seen or 72.25 and 68.921 **[M1]**

(ii) 1.49 **[B1]**

(b) 96 **[B2]**

If 0 scored, M1 for 23.04 or 0.24 seen **[M1]**

[Back to Question] *[Total Marks: 5]*

15 (a) $\sqrt{151.83}$ or 12.32... or 29.791 **[M1]**

0.4136... **[A1]**

(b) $p \times q = 3402$ **[M1]**

$\sqrt[3]{3402}$ or 15.0401... **[M1]**

15.0 **[A1]**

[Back to Question] *[Total Marks: 5]*

16 **(a)** 1.59 **[B2]**

M1 for 23.02... or 14.44 seen.

(b) 1734 **[B3]**

M1 for $\sqrt[3]{4913}$ or 17 seen.

M1 for 6×17^2.

[Back to Question] *[Total Marks: 5]*

17 **(a)** 1.5 **[B1]**

(b) Rearranges to $2x^4 = 1250$ leading to $x^4 = 625$ **[M1]**

$x = 5$ **[B1]**

$x = -5$ **[B1]**

(c) Correct substitution: $\sqrt{\dfrac{3(4.2)^2 + (-5.7)}{1.8}}$ **[M1]**

Evaluates numerator to 47.22 OR fraction to 26.2333... **[M1]**

5.12 **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** 0.239 **[B2]**

If 0 scored, allow [M1] for evaluating the numerator as 19.679... or the denominator as 82.464...

(b) 64 **[B3]**

If 0 scored, allow [M1] for isolating x^2 correctly, e.g., $x^2 = 16^3$ or $x^2 = 4096$

Allow a further [M1] for $x = \sqrt{4096}$

(c) 7.44 or 7.443... **[B3]**

If 0 scored, allow [M1] for finding the volume of the cube as 12^3 or 1728

Allow a further [M1] for correctly rearranging the sphere volume formula to solve for r, e.g., $r^3 = \dfrac{1728 \times 3}{4\pi}$ or $r = \sqrt[3]{\dfrac{1296}{\pi}}$

[Back to Question] *[Total Marks: 8]*